Logic Minimization– Part 1

K. Sridharan

Review

- We have discussed algebraic simplification of Boolean functions.
- We will discuss more about minterms and maxterms in this lecture.

Terminology

- When a function f is written as the sum of products, it is also called minterm expansion. The minterm expansion of a Boolean function f is unique.
- We also saw one can express f as the product of maxterms. This is called maxterm expansion.
- We observed that one can derive all minterms for a Boolean function using the truth table. Can one do this without a truth table ?

Can we derive all minterms without a truth table?

- Question for students.

Finding minterm expansion without a truth table

- Minterm expansion for f = a'(b' + d) + acd'

$$
\begin{aligned}
f &= a'b' + a'd + acd' \\
&= a'b'(c + c')(d + d') + a'd(b + b')(c + c') + acd'(b + b') \\
&= a'b'c'd' + a'b'c'd + a'b'cd' + a'b'cd + a'b'c'd + a'b'cd \\
&\quad + a'bc'd + a'bcd + abcd' + ab'cd'
\end{aligned}
$$

a'b'cd and a'b'c'd are appearing twice so can crossout one occurrence

Finding the maxterm expansion

- Once again the truth table is not explicitly required.
- For the example in the previous slide, simply write out the decimal integers corresponding to the minterms.
- Then list the decimal integers which do not correspond to the minterms and write out the maxterm expansion. Example next.

Example for maxterm expansion

- Consider the minterm expansion of
  f = a'(b' +d) + acd'.
- This corresponds to $\sum$ (0,1,2,3,5,7,10,14)
  (see the minterm expansion slide
  earlier starting with a'b'c'd')
- Hence, the "left out" terms are 4,6,8,9,11,12,13 and 15. The maxterm expansion is therefore
  $\prod$ (4,6,8,9,11,12,13,15). We have only four
  variables hence the maximum decimal no. is 15

Any other use of minterm expansion ?

Can show equivalence of two expressions

Show a'c + b'c'+ ab = a'b' + bc + ac'

Exercise for students.

Any other use of minterm expansion ?

Can show equivalence of two expressions

Show a'c + b'c'+ ab = a'b' + bc + ac'

For the left hand side, we have

$a'c(b+b') + b'c'(a+a') + ab(c+c')$

$= a'bc + a'b'c + ab'c' + a'b'c' + abc + abc'$

$= \Sigma(3,1,4,0,7,6) \rightarrow (I)$

For the right hand side, we have

$a'b'(c+c') + bc(a+a') + ac'(b+b')$

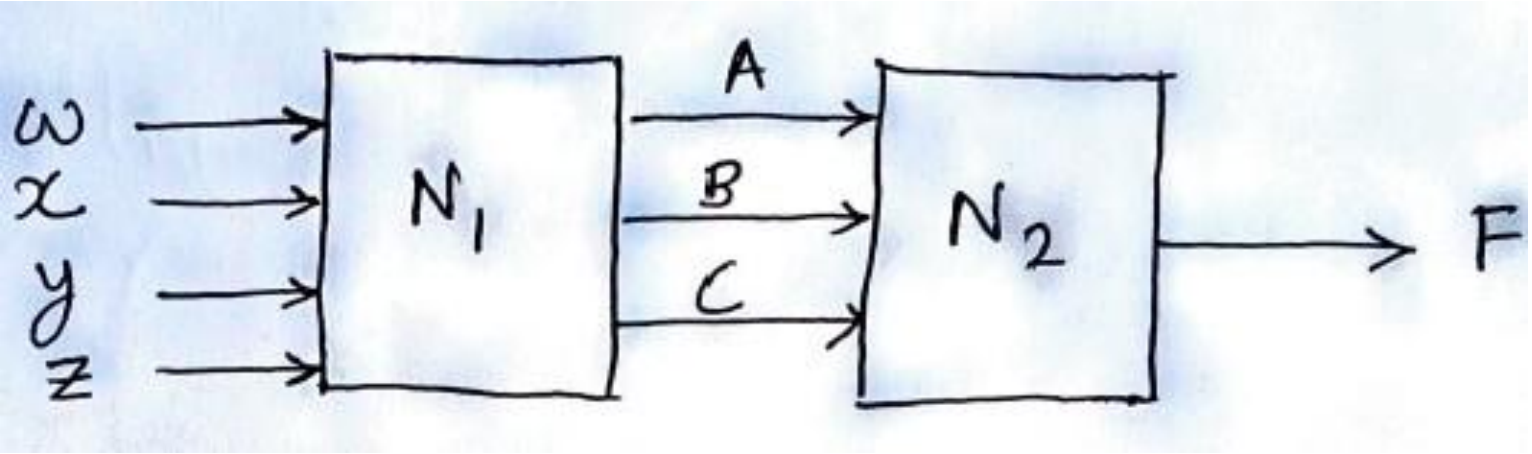
$= a'b'c + a'b'c' + abc + a'bc + abc' + ab'c'$

$= \Sigma(1,0,7,3,6,4) \rightarrow (II)$

Since (I) = (II), the given equation is valid.

Incompletely specified functions

- We now move on to another aspect with regard to digital systems. Consider the figure shown below:



Explanation

- We consider a digital system composed of two subnetworks.
- Let the output of network N_1 drive the input of network N_2 and let us assume that the output of N_1 does not generate all possible combinations of values for A, B and C.
- In particular, we'll assume that there are no combinations of w,x, y and z that will cause A,B and C to have values of 001 or 110.

How will this reflect in the truth table ?

When we design N_2 , we need not specify values of F for ABC=001 or 110 (these combinations will not occur as inputs to N_2)

A	B	C	F
0	0	0	1
0	0	1	X
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	X
1	1	1	1

X is called a don't care; F is therefore incompletely specified

What is the consequence of an incompletely specified function ?

- It helps in getting a “simplified” solution during realization.
- When we realize the function, we need to specify values for the don’t cares (note the difference between design and realization; we spoke about design of N_2 in the previous slide).
- We can see how the function F will look like for different assignments to X (don’t care) and then choose the best (i.e., the one that gives the most simplified form for F)

What happens in this example ?

- If we assign the value 0 to both X's, then we get F = A'B'C' + BC
- If we assign 1 to the first X and 0 to the second, we get F = A'B' + BC
- If we assign 1 to both X's, we get

  F = A'B' + BC + AB
- Hence, the choice of 1 to the first X and 0 to the second is the best (shown in red).

Other possibilities for incompletely specified functions

- Some other scenarios are also possible for incompletely specified functions.
- For instance, we can have all input combinations for network N_2 but the network output may be used in such a way that we don't care whether it is 1 or 0 for some input combinations.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021.